Correlation Dynamics and Potential Functions of Market States

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In present work we investigate the non stationarity of the mean correlation coefficient *c* of SP500 stocks in a time period of 21 years. Modeling *c* as a stochastic process we extract deterministic dynamics directly form the empirical times series using known methods [1, 2]. We find a universal and time independent as well as strongly non stationary parts in its deterministic dynamics.

We additionally apply our estimation methods to artificial time series showing the accuracy of the estimation methods.

Furthermore we use these methods to analyze properties of the market states (as introduced in [3]). Through the shape of the potential functions of the deterministic forces we are able to discuss the stability of the market states, the state transition as well as their hierarchical structure.

References:

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